

The Geometric Part of Decoherence: Quasi-Orthogonality in High-Dimensional Hilbert Spaces

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We isolate a geometric mechanism that complements the dynamical suppression of macroscopic interference: In a high-dimensional Hilbert space, almost all state vectors are nearly orthogonal, accommodating an exponentially large reservoir of mutually quasi-orthogonal environmental records. This geometry explains why macroscopic alternatives fail to exhibit visible interference once such records are populated. The argument is conditional and finite-dimensional, and it leaves the interpretive core of quantum mechanics untouched: geometry alone does not select a pointer basis, does not guarantee that a given Hamiltonian drives the system into typical regions of the accessible subspace, and does not turn an improper mixture into a proper one. It merely supplies the vast Hilbert-space capacity that makes decoherence so overwhelmingly effective for all practical purposes.

I. INTRODUCTION

The quantum measurement problem asks how the continuous and reversible Schrödinger evolution gives rise to the discrete, irreversible classical world we observe [1, 2]. Operationally, the question reduces to why the off-diagonal interference terms of a system’s reduced density matrix are unobservable on macroscopic scales.

The urgency of this question was forcefully articulated by Schrödinger in a 1952 colloquium [3]. He noted that if the wave equation is universally valid, then alternative macroscopic outcomes must all “really happen simultaneously.” Schrödinger argued that if nature actually took this form, we should find our surroundings rapidly blurring together into a “quagmire, or sort of a featureless jelly”—a variant of his cat metaphor. For example, a watch left unobserved in a drawer overnight should dissolve into a smeared probability distribution. This “jellification” argument poses a sharp, physical question: if unitary quantum mechanics holds at all scales, what fundamental mechanism prevents the simultaneous alternatives of the macroscopic world from blurring into an unobservable, featureless smear?

Bell coined “FAPP” (*for all practical purposes*) to express skepticism toward explanations that rest on the practical impossibility of observing macroscopic interference rather than on a fundamental physical mechanism [4]. Modern decoherence theory [1, 5–7] answers this by showing that interaction with a macroscopic environment dynamically drives the off-diagonal terms toward zero in a preferred (pointer) basis selected by the system–environment coupling.

We shall examine a complementary, purely *geometric* contribution to the same suppression. The relevant tools are the exponential packing of approximately orthogonal unit vectors and the concentration of measure on high-dimensional spheres [8–12]. These results are standard in

high-dimensional geometry, but their direct application to the environmental records appearing in decoherence is, to our knowledge, underemphasized.

Our central claim is modest. High-dimensional Hilbert-space geometry guarantees an enormous reservoir of nearly non-interfering pure states. This is a kinematic fact independent of any Hamiltonian. Whether actual environmental branches occupy this reservoir is a separate dynamical question. In ordinary decoherence theory, the system–environment interaction selects a pointer basis and correlates different pointer values with different environmental records. The geometric estimates below apply once those records are sufficiently typical in the accessible environmental subspace.

We therefore separate two ingredients that are often bundled together. The dynamical ingredient is the production of environmental records by the Hamiltonian. The kinematic ingredient is the concentration of measure on high-dimensional spheres, which makes generic records almost mutually orthogonal. The latter sharply quantifies the FAPP suppression of macroscopic interference, while leaving untouched the deeper interpretive question of whether an improper mixture has become a proper one.

II. QUASI-ORTHOGONALITY AND EXPONENTIAL PACKING

In a d -dimensional Hilbert space $\mathcal{H}_d$, the maximum number of exactly mutually orthogonal unit vectors is d . We relax exact orthogonality as follows. Two unit vectors $|\psi_i\rangle, |\psi_j\rangle \in \mathcal{H}_d$ are called ε -quasi-orthogonal if

$$|\langle\psi_i|\psi_j\rangle|^2 \leq \varepsilon. \quad (1)$$

A family $\{|\psi_i\rangle\}_{i=1}^M$ is ε -quasi-orthogonal if this inequality holds for every pair $i \neq j$.

Let $M_\varepsilon(d)$ denote the largest possible cardinality of such a family. The relevant fact for us is the following

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random-coding lower bound:

$$M_\varepsilon(d) \geq \left\lfloor \exp \left[\frac{d-1}{2} (-\log(1-\varepsilon)) - \frac{1}{2} \right] \right\rfloor. \quad (2)$$

For small fixed ε , since $-\log(1-\varepsilon) = \varepsilon + O(\varepsilon^2)$, this implies

$$M_\varepsilon(d) \geq \exp(c\varepsilon d) \quad (3)$$

for some constant $c > 0$, fixed $\varepsilon > 0$, and sufficiently large d . The proof is given in Appendix A. The statement should be understood in the fixed- ε , large- d regime; no bound of the form $M_\varepsilon(d) \leq \exp(C\varepsilon d)$ can hold uniformly down to $\varepsilon = O(1/d)$, because an orthonormal basis already gives $M_\varepsilon(d) \geq d$ for every $\varepsilon \geq 0$. Standard spherical-code bounds also give exponential-in- d upper bounds for fixed $\varepsilon > 0$ [13, 14], but the precise optimal exponent is not needed here. We therefore use only the random-coding lower bound.

For a system of n qubits, $d = 2^n$. Hence, for fixed $\varepsilon > 0$,

$$M_\varepsilon(2^n) \geq \exp(c\varepsilon 2^n), \quad (4)$$

which is doubly exponential in the number n of qubits. Thus the Hilbert space of a macroscopic environment has vastly more than enough kinematic room to accommodate an enormous number of mutually quasi-orthogonal environmental records.

III. LÉVY'S LEMMA AND TYPICAL OVERLAPS

The packing bound concerns the existence of large quasi-orthogonal families. A complementary statement is that quasi-orthogonality is also typical: two independently chosen random pure states in a high-dimensional Hilbert space almost certainly have very small overlap.

The basic reason is geometric. In high dimension, most of the volume of the unit sphere lies close to the equator relative to any fixed direction. Equivalently, if one fixes a reference state $|\phi\rangle$, then a Haar-random state $|\psi\rangle$ has only a very small projection onto $|\phi\rangle$ with overwhelming probability. Lévy's lemma gives one general concentration-of-measure formulation of this fact.

We first fix the terminology. Identify $\mathcal{H}_d \cong \mathbb{C}^d \cong \mathbb{R}^{2d}$, and let

$$\mathbb{S}^{2d-1} = \{|\psi\rangle \in \mathbb{C}^d : \langle\psi|\psi\rangle = 1\} \quad (5)$$

be the unit sphere. The probability measure used below is the uniform measure on this sphere. Equivalently, it is the measure induced by Haar measure on the unitary group $U(d)$: if $|\psi\rangle$ is distributed according to this measure, then $U|\psi\rangle$ has the same distribution for every unitary $U \in U(d)$. We write $|\psi\rangle \sim \mu_{\text{Haar}}$ and call such a vector Haar-random.

Throughout the paper, $\mathbb{E}$ denotes expectation value, i.e. an average over the probability distribution under discussion, and $\mathbb{P}$ denotes the corresponding probability of an event. When no measure is specified explicitly, the relevant distribution is the Haar distribution on the unit sphere.

The metric on $\mathbb{S}^{2d-1}$ will be the Euclidean, or chordal, distance inherited from $\mathbb{C}^d$:

$$\|\psi - \varphi\| = \sqrt{\langle\psi - \varphi|\psi - \varphi\rangle}. \quad (6)$$

A function $f : \mathbb{S}^{2d-1} \rightarrow \mathbb{R}$ is called L -Lipschitz if

$$|f(\psi) - f(\varphi)| \leq L \|\psi - \varphi\| \quad (7)$$

for all unit vectors $|\psi\rangle, |\varphi\rangle$. The smallest such L is the Lipschitz constant of f . Intuitively, L measures how rapidly f can change when its input vector is moved on the unit sphere.

Lévy's lemma [11, 12, 15] is a concentration-of-measure theorem. In one convenient normalization, it states that if $f : \mathbb{S}^{2d-1} \rightarrow \mathbb{R}$ is L -Lipschitz, then for every $\delta > 0$,

$$\mathbb{P}_{\mu_{\text{Haar}}} (|f(\psi) - \mathbb{E}_{\mu_{\text{Haar}}} f| \geq \delta) \leq 2 \exp \left[-\frac{(2d-1)\delta^2}{9\pi^3 L^2} \right]. \quad (8)$$

Here

$$\mathbb{E}_{\mu_{\text{Haar}}} f = \int_{\mathbb{S}^{2d-1}} f(\psi) d\mu_{\text{Haar}}(\psi) \quad (9)$$

is the average value of f over the unit sphere. Some versions of Lévy's lemma are stated using a median of f rather than its mean; the two forms differ only by inessential numerical constants.

The important point is the dependence on d : for any observable f that is not too sensitive to small changes of the state vector, almost all states on a high-dimensional sphere give values of f extremely close to the average value. The phrase “sub-Gaussian tail” means that the probability of a deviation of size δ is bounded by an expression of the form $\exp[-\text{const.} \times d\delta^2]$, just as for a Gaussian random variable whose variance shrinks like $1/d$.

This statement by itself says only that a smooth function is almost constant on most of the sphere. To obtain quasi-orthogonality, one must apply it to a function whose value measures non-orthogonality. The relevant function is the squared overlap with a fixed state. Its concentration near its mean implies quasi-orthogonality precisely because that mean is itself small, of order $1/d$.

We now apply this general theorem to the squared overlap with a fixed reference state. Fix $|\phi\rangle \in \mathcal{H}_d$ with $\langle\phi|\phi\rangle = 1$, and define

$$f_\phi(\psi) = |\langle\phi|\psi\rangle|^2. \quad (10)$$

This is a real-valued function on $\mathbb{S}^{2d-1}$. It directly measures the amount of $|\psi\rangle$ lying in the direction $|\phi\rangle$. If $f_\phi(\psi) = 1$, the two states are identical up to a phase; if

$f_\phi(\psi) = 0$, they are exactly orthogonal; and if $f_\phi(\psi) \ll 1$, they are nearly orthogonal.

Its Haar mean is

$$\mathbb{E}_{|\psi\rangle \sim \mu_{\text{Haar}}} |\langle \phi | \psi \rangle|^2 = \frac{1}{d}. \quad (11)$$

Indeed, by unitary invariance we may take $|\phi\rangle = |e_1\rangle$ in an orthonormal basis $\{|e_k\rangle\}_{k=1}^d$. Since

$$\sum_{k=1}^d |\langle e_k | \psi \rangle|^2 = 1, \quad (12)$$

and all d terms have the same expectation by symmetry, each has expectation $1/d$.

Equation (11) is the point at which concentration becomes quasi-orthogonality. Lévy's lemma will say that $f_\phi(\psi)$ is typically close to its average. But the average value is $1/d$, which is tiny for large d . Therefore a typical Haar-random state has squared overlap of order $1/d$ with the fixed state $|\phi\rangle$, and hence overlap amplitude of order $1/\sqrt{d}$:

$$|\langle \phi | \psi \rangle|^2 \sim \frac{1}{d}, \quad |\langle \phi | \psi \rangle| \sim \frac{1}{\sqrt{d}}. \quad (13)$$

Thus “almost constant” means “almost constantly close to zero” for this particular observable.

The function f_ϕ is also Lipschitz. For unit vectors $|\psi\rangle, |\varphi\rangle$,

$$||\langle \phi | \psi \rangle|^2 - |\langle \phi | \varphi \rangle|^2| \quad (14)$$

$$\begin{aligned} &= |(|\langle \phi | \psi \rangle| - |\langle \phi | \varphi \rangle|)(|\langle \phi | \psi \rangle| + |\langle \phi | \varphi \rangle|)| \\ &\leq (|\langle \phi | \psi \rangle| + |\langle \phi | \varphi \rangle|) |\langle \phi | \psi - \varphi \rangle| \\ &\leq 2 \|\psi - \varphi\|. \end{aligned} \quad (15)$$

Thus f_ϕ is L -Lipschitz with $L \leq 2$. Substituting $L = 2$ and Eq. (11) into Lévy's lemma gives

$$\mathbb{P}\left(\left||\langle \phi | \psi \rangle|^2 - \frac{1}{d}\right| \geq \delta\right) \leq 2 \exp\left[-\frac{(2d-1)\delta^2}{36\pi^3}\right]. \quad (16)$$

In particular,

$$\mathbb{P}\left(|\langle \phi | \psi \rangle|^2 \geq \frac{1}{d} + \delta\right) \leq 2 \exp\left[-\frac{(2d-1)\delta^2}{36\pi^3}\right]. \quad (17)$$

This inequality says that significant excess overlap with the fixed state $|\phi\rangle$ is exponentially unlikely in the Hilbert-space dimension. Geometrically, the set of states having appreciable overlap with $|\phi\rangle$ is a small cap around the $|\phi\rangle$ direction, while most of the high-dimensional sphere lies close to the equator orthogonal to $|\phi\rangle$.

For the particular random variable

$$X = |\langle \phi | \psi \rangle|^2, \quad (18)$$

one can do better than Lévy's lemma. By unitary invariance, again take $|\phi\rangle = |e_1\rangle$. A Haar-random vector can

be generated by choosing independent standard complex Gaussian variables $g_1, \dots, g_d$ and normalizing:

$$|\psi\rangle = \frac{1}{\sqrt{\sum_{k=1}^d |g_k|^2}} \sum_{k=1}^d g_k |e_k\rangle. \quad (19)$$

Then

$$X = \frac{|g_1|^2}{\sum_{k=1}^d |g_k|^2}. \quad (20)$$

The variables $|g_k|^2$ are independent exponential random variables. It follows that X has the Beta distribution $\text{Beta}(1, d-1)$, whose density is

$$p_d(x) = (d-1)(1-x)^{d-2}, \quad 0 \leq x \leq 1. \quad (21)$$

Therefore the exact survival probability is

$$\begin{aligned} \mathbb{P}(X \geq \epsilon) &= \int_\epsilon^1 (d-1)(1-x)^{d-2} dx \\ &= (1-\epsilon)^{d-1} \leq e^{-(d-1)\epsilon}. \end{aligned} \quad (22)$$

Thus, for any fixed threshold $\epsilon > 0$, the probability that a Haar-random state has squared overlap at least ϵ with $|\phi\rangle$ decays exponentially in d . This sharper calculation makes the same point as Lévy's lemma: large overlaps occupy exponentially little measure on the high-dimensional unit sphere.

Equation (22) is sharper than the general Lévy bound for this particular observable. Lévy's lemma is nevertheless useful because it exhibits the general geometric mechanism: in high dimension, any sufficiently regular real-valued function on the unit sphere is almost constant on almost all of the sphere. Applied to overlaps, this does not merely say that the overlap function is concentrated; it says that it is concentrated near its mean value $1/d$. Since $1/d$ is small in a large Hilbert space, a Haar-random state is almost certainly nearly orthogonal to any fixed state.

Finally, the same conclusion holds for two independently Haar-random states. If $|\psi\rangle$ and $|\varphi\rangle$ are independently distributed according to μ_{Haar} , then one may condition on $|\varphi\rangle$. For that fixed value of $|\varphi\rangle$, the preceding argument applies to

$$|\langle \varphi | \psi \rangle|^2. \quad (23)$$

Averaging over $|\varphi\rangle$ then gives the same distribution as in Eq. (22). Hence two independently chosen pure states in a large Hilbert space are quasi-orthogonal with overwhelming probability.

IV. APPLICATION TO THE MEASUREMENT PROCESS

Consider a system $\mathcal{S}$ coupled to an apparatus-environment $\mathcal{E}$ initialised in $|E_0\rangle$:

$$|\Psi(0)\rangle = \left(\sum_i c_i |s_i\rangle\right) \otimes |E_0\rangle. \quad (24)$$

A measurement-like unitary correlates pointer outcomes with environmental records:

$$|\Psi(t)\rangle = \sum_i c_i |s_i\rangle \otimes |E_i(t)\rangle. \quad (25)$$

The reduced density matrix on $\mathcal{S}$ is

$$\rho_{\mathcal{S}}(t) = \sum_{i,j} c_i c_j^* \langle E_j(t) | E_i(t) \rangle |s_i\rangle \langle s_j|. \quad (26)$$

Off-diagonal coherences are weighted by the environmental overlaps $\langle E_j(t) | E_i(t) \rangle$, and are suppressed in modulus by at most $\sqrt{\varepsilon}$ whenever the branch states are ε -quasi-orthogonal.

The kinematic claim.

Throughout this section we adopt natural units $k_B = \hbar = 1$, so that entropies are dimensionless and inverse temperatures have units of time. If the branch states $\{|E_i(t)\rangle\}_{i \neq j}$ are well-modeled as a typical pair on the relevant high-dimensional subspace of $\mathcal{H}_{\mathcal{E}}$, then

$$\mathbb{E}[|\langle E_j(t) | E_i(t) \rangle|^2] = \frac{1}{d_{\text{eff}}}, \quad (27)$$

with d_{eff} the dimension of the subspace explored by the dynamics. For a generic chaotic many-body environment the accessible subspace is the microcanonical shell, so

$$d_{\text{eff}} \sim e^S, \quad S \sim n, \quad (28)$$

with S the (dimensionless) microcanonical entropy of the relevant environmental sector and n the number of constituent degrees of freedom; in conventional units the right-hand sides of (28) read e^{S/k_B} and nk_B , respectively (see also Appendix B). Equation (28) ties the geometric parameter d_{eff} directly to standard thermodynamics: the mean-square overlap is exponential in the entropy of the environment, so the typical overlap amplitude is of order $e^{-S/2}$. This is vastly stronger than needed to render coherences operationally invisible on any timescale shorter than the Poincaré recurrence time, which for a generic chaotic many-body system is astronomically large—heuristic arguments suggest a scaling at least as $\exp(d_{\text{eff}}) \sim \exp(e^S)$, but the precise value is model-dependent and irrelevant to our argument [16, 17].

Minimal dynamical input.

The estimates above are kinematic. They apply to actual environmental branches only if the branch states generated by the Hamiltonian are sufficiently typical in the accessible environmental subspace. In a measurement-like interaction, after a pointer basis has been selected, one may write schematically

$$|E_i(t)\rangle = U_i(t)|E_0\rangle, \quad (29)$$

where the conditional unitary $U_i(t)$ depends on the pointer value i . The geometric argument does not prove that the vectors $U_i(t)|E_0\rangle$ are Haar-random. The required dynamical assumption is only the weaker pair-typicality condition that, for $i \neq j$, their overlap statistics are close to those of two typical vectors in the accessible subspace $\mathcal{H}_{\text{acc}}$:

$$|\langle E_j(t) | E_i(t) \rangle|^2 \sim d_{\text{eff}}^{-1}, \quad d_{\text{eff}} = \dim \mathcal{H}_{\text{acc}}. \quad (30)$$

Chaotic, nonintegrable environments provide the usual physical reason to expect such typicality after a suitable relaxation or decoherence time. Integrable, many-body-localized, weakly coupled, or strongly constrained environments may fail to satisfy it. Thus the geometry supplies the available reservoir of nearly orthogonal records, while the Hamiltonian must still populate that reservoir. This assumption is in the same broad family as the typicality assumptions used in canonical typicality and quantum thermalization [18–22].

Limitations.

Two limitations should be kept explicit. First, quasi-orthogonality does not select the pointer basis. The preferred basis is determined by the structure of the system–environment interaction, not by high-dimensional geometry alone.

Second, small off-diagonal terms do not by themselves turn an improper mixture into a proper one. The geometric estimates therefore strengthen the FAPP suppression of interference, but they do not resolve the interpretive core of the measurement problem.

A strict, rather than merely FAPP, breakdown of unitary equivalence between macroscopic outcomes can be obtained in the infinite tensor product limit through sectorization and factorization [23–26], of which the present finite-dimensional, geometric estimates may be viewed as a quantitative companion: exponentially small (rather than exactly vanishing) overlaps between macroscopically distinct environmental records.

Schrödinger’s “jellification” worry.

Schrödinger’s worry was that, if unitary quantum mechanics applies universally, macroscopically distinct alternatives should blur into a single “featureless jelly.” In the present language, such blurring would require appreciable overlaps between the environmental records correlated with different macroscopic alternatives. Under the pair-typicality assumption above, however,

$$|\langle E_j(t) | E_i(t) \rangle|^2 \sim d_{\text{eff}}^{-1} \sim e^{-S},$$

so the typical overlap amplitude is of order $e^{-S/2}$. For a macroscopic environment this is FAPP indistinguishable from zero. The branches do not blur because the

accessible Hilbert space contains an enormous supply of nearly orthogonal directions. This answers the jellification worry operationally, while leaving the deeper interpretive questions untouched.

V. DISCUSSION

The geometric perspective separates two ingredients in decoherence. The first is dynamical: the interaction Hamiltonian selects a pointer basis and correlates distinct pointer values with distinct environmental records. The second is kinematic: once those records behave like typical vectors in a high-dimensional accessible subspace, their mutual overlaps are automatically tiny.

For an accessible subspace of dimension $d_{\text{eff}} \sim e^S$, the typical squared overlap is $d_{\text{eff}}^{-1} \sim e^{-S}$, and the typical overlap amplitude is $e^{-S/2}$. This suppression is not a new dynamical mechanism; it is the high-dimensional geometry underlying the effectiveness of environmental decoherence. The geometry explains why there is ample Hilbert-space room for many macroscopic records to coexist without appreciable interference, but it does not explain why a particular Hamiltonian produces those records, nor does it select a preferred basis.

This distinction also clarifies where the framework can fail. If the environment is integrable, many-body localized, weakly coupled, or restricted by additional conserved quantities, the branch states may not be typical in the relevant subspace. In that case the geometric reservoir exists but is not dynamically populated.

The argument is close in spirit to canonical typicality, where concentration of measure is used to show that most pure states of a large composite system have nearly thermal reduced states on small subsystems [18–20]. Here the object of interest is different: the pairwise overlap between distinct environmental records. Infinite-tensor-product approaches to measurement obtain exact orthogonality between macroscopic sectors in the limit $n = \infty$; the present analysis can be read as the finite- n version, where exact sectorization is replaced by exponentially small overlap.

VI. CONCLUSION

We have isolated a kinematic contribution to the suppression of macroscopic interference. In a d -dimensional Hilbert space, random pure states are almost orthogonal: for fixed $|\phi\rangle$ and Haar-random $|\psi\rangle$,

$$\mathbb{E}|\langle\phi|\psi\rangle|^2 = \frac{1}{d}, \quad \mathbb{P}(|\langle\phi|\psi\rangle|^2 \geq \varepsilon) = (1 - \varepsilon)^{d-1}.$$

Moreover, for fixed $\varepsilon > 0$, random coding produces ε -quasi-orthogonal families of size at least $\exp(c\varepsilon d)$. For an environmental subspace with $d_{\text{eff}} \sim e^S$, this means that typical squared branch overlaps are of order e^{-S} .

These facts do not replace decoherence dynamics. The Hamiltonian must still select the pointer basis and drive environmental records into typical regions of the accessible subspace. Nor do the estimates convert an improper mixture into a proper one. What the geometry does provide is a quantitative explanation of why, once environmental records have formed, macroscopic alternatives can coexist without producing Schrödinger’s “featureless jelly”: high-dimensional Hilbert space supplies an enormous reservoir of nearly non-interfering directions.

Appendix A: Lower Bound on Quasi-Orthogonal Packing

Probabilistic construction. Sample M unit vectors $|\psi_1\rangle, \dots, |\psi_M\rangle \in \mathcal{H}_d$ independently from the Haar measure on $\mathbb{S}^{2d-1}$. For any pair $i \neq j$, by the exact Beta tail (22),

$$\mathbb{P}(|\langle\psi_i|\psi_j\rangle|^2 \geq \varepsilon) = (1 - \varepsilon)^{d-1}.$$

By a union bound over the $\binom{M}{2} < M^2/2$ pairs,

$$\mathbb{P}(\exists i \neq j : |\langle\psi_i|\psi_j\rangle|^2 \geq \varepsilon) < \frac{1}{2}M^2(1 - \varepsilon)^{d-1}.$$

Choosing $M = \lfloor \exp[\frac{d-1}{2}(-\log(1 - \varepsilon)) - \frac{1}{2}] \rfloor$ makes this bound strictly less than 1, so an ε -quasi-orthogonal set of size M exists. $\square$

This proves the lower bound used in the main text. Standard spherical-code estimates give exponential-in- d upper bounds for fixed $\varepsilon > 0$, but the precise optimal exponent is not needed here. One should not state a uniform upper bound of the form $\exp(C\varepsilon d)$ down to $\varepsilon = O(1/d)$, since an orthonormal basis already gives $M_\varepsilon(d) \geq d$ for every $\varepsilon \geq 0$.

Appendix B: Effective Dimension and the Microcanonical Entropy

In (16) the relevant dimension is the dimension of the subspace $\mathcal{H}_{\text{acc}} \subset \mathcal{H}_{\mathcal{E}}$ accessible to the dynamics from the initial state on the timescale considered, not the full Hilbert space dimension. Energy conservation, conserved charges, and finite-time evolution all restrict the accessible subspace. Concretely, for a many-body system with Hamiltonian $H_{\mathcal{E}}$ and an initial state localized within an energy window $[E, E + \Delta E]$, the accessible subspace is well approximated by the microcanonical shell, and its dimension is

$$d_{\text{eff}} = \dim \mathcal{H}_{\text{acc}} \sim e^{S(E)/k_B} \quad (\text{B1})$$

with $S(E)$ the microcanonical entropy. In the natural units adopted in Sec. IV ($k_B = \hbar = 1$), the right-hand side reads $e^{S(E)}$, consistent with (28). For chaotic many-body dynamics with bounded local interactions, $S(E)$ is

extensive in n (Boltzmann scaling), so d_{eff} remains exponential in n and the qualitative conclusions of Sec. IV survive.

For integrable or many-body-localized systems, conserved local quantities restrict the dynamics to a polynomial-in- n submanifold of the microcanonical shell [21, 22], and d_{eff} can be polynomial or even constant in n . The geometric reservoir is correspondingly smaller, and the conditional dynamical assumption of Sec. IV is correspondingly violated. This is a further reason to view the geometric statement as conditional on a dynamical input, and provides a quantitative diagnostic: any ex-

perimental observation of slower-than- e^{-S} decoherence is direct evidence of obstructions to effective typicality on the accessible subspace.

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